

# LinkedIn Job-Change Lead Engine — Requirements

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## Summary and goals

The system turns public job-change announcements into a ranked, mapped list of likely movers, because starting a new job — especially in a new metro — is one of the strongest leading indicators that someone will buy, sell or rent within 12 months.

End to end, it does four things:

1. **Acquire** posts that announce a job change, scoped to one or more target markets.
2. **Extract** structured fields from the free text: person, new title, new employer, new location, start date, and whether the move is a relocation.
3. **Store** each lead with its provenance, confidence and score in a queryable database.
4. **Surface** leads to an agent in a dashboard — a scrollable card feed and a map — so they can decide who to call today.

**Primary user.** A residential real estate agent or inside-sales associate working a defined geographic farm area. They open the dashboard each morning and need one answer: *which five people should I reach out to today, and what do I say?*

**Secondary user.** A team lead or brokerage admin who assigns leads, tunes the scoring, sets the service area and reviews data quality.

**Success looks like** an agent trusting the feed enough to work it daily without re-verifying each lead by hand, and the first closed transaction traceable to a lead the system found.

## Legal and compliance constraints

Read this before the technical sections, because it changes what you build, not just how you operate it. Four separate bodies of rule apply at once, and three of them survive any clever engineering.

**LinkedIn's User Agreement prohibits automated scraping.** This is a contract question, not a criminal one. The Ninth Circuit's decision in *hiQ Labs v. LinkedIn* held that scraping publicly visible data is not "unauthorized access" under the Computer Fraud and Abuse

Act — but hiQ subsequently lost on breach of contract and the matter resolved with hiQ enjoined from scraping. The practical reading: automated collection is unlikely to be a crime and is still a breach of contract that LinkedIn enforces, with account bans and injunctions as the live risks.

**Once you store the data, you are a data controller.** Names, employers, locations and an inferred intent to move are personal data. Under CCPA/CPRA and GDPR that creates obligations you have to design for: disclosure of what you hold, deletion on request, and an opt out of sale or sharing. Small brokerages may fall below CCPA's revenue and volume thresholds today, but the deletion and suppression machinery is far cheaper to build in now than to retrofit.

**The Fair Housing Act is the one most teams miss.** Targeting real estate marketing by geography, and scoring prospects, can become unlawful steering or discrimination when the model uses protected characteristics or proxies for them. Race, color, religion, sex, familial status, national origin and disability are off limits — and so are the things that stand in for them: name-derived ethnicity, profile photos, alma mater, and neighborhood demographic data. The scoring model must be inspectable and its inputs logged.

**Outreach has its own rules.** The TCPA governs calls and texts (consent requirements for autodialed or prerecorded contact, plus Do Not Call scrubbing), CAN-SPAM governs email (working opt out, physical address, honest headers), and your state's real estate advertising and licensing rules apply on top.

## What this means for the build

- The data source must be a **swappable adapter**, not a hardcoded scraper. If the chosen source becomes untenable, you replace one module.
- Every lead carries **provenance**: which source, which URL, fetched when, under which terms.
- A **suppression list** and a hard-delete path are MVP scope, not v2 scope.
- The scoring model excludes protected-class fields and proxies by design, and logs its inputs for audit.
- Geocoding resolves to a **city or metro centroid**, never an inferred home address.

None of this is legal advice. Have counsel review the data source choice and the outreach flow before you put the system in front of customers — the source decision in particular is cheap to change now and expensive later.

## Data sources and acquisition

Treat the source as a pluggable decision, then pick the lowest-risk option that produces enough volume for your market. Direct scraping is the obvious path and the worst one to build your only pipeline on.

| Option                              | What it gives you                                             | Volume                            | Legal risk                                     | Notes                                                                                                         |
|-------------------------------------|---------------------------------------------------------------|-----------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Sales Navigator job-change alerts   | LinkedIn's own "changed jobs" signal on saved leads           | Low, and only for lists you build | Lowest — a supported product feature           | Requires a seat per user; no API for most tiers, so capture is manual or extension-assisted                   |
| Licensed people-data providers      | Bulk person records with employment history and change deltas | High                              | Low — contractual, with indemnities            | Costs real money; vendor availability shifts as LinkedIn litigates, so evaluate who is actually selling today |
| "People on the Move" business press | Job changes published explicitly for promotion                | Moderate, skews senior            | None — published for this purpose              | Local business journals, press releases, company newsroom pages. Genuinely underused                          |
| User-session browser extension      | Posts the agent already sees in their own authenticated feed  | Low to moderate                   | Medium — human in the loop, still ToS-adjacent | Slow to accumulate; strongest story is "it helps me read my own feed"                                         |

| Option                    | What it gives you          | Volume                        | Legal risk                                     | Notes                                                                         |
|---------------------------|----------------------------|-------------------------------|------------------------------------------------|-------------------------------------------------------------------------------|
| LinkedIn official APIs    | Marketing and partner data | Effectively none for this use | Lowest                                         | Be clear-eyed: there is no sanctioned post firehose. Do not plan around one   |
| Direct automated scraping | Everything you want        | High                          | Highest — breach of contract, bans, injunction | Works until it doesn't, and it fails at the worst time                        |
| Manual and CSV import     | Whatever a human pastes in | Low                           | None                                           | Always support it — it is your test harness, your fallback and your demo path |

**Recommendation.** Build the pipeline against licensed data plus the business-press feed, and keep CSV import as the always-available path. Those three cover the thesis without betting the company on a source that can be switched off by someone else's legal team.

## Source adapter requirements

- **SRC-1** Every source implements one `SourceAdapter` interface: `discover(query, cursor)` returns candidate items plus a next cursor; `fetch(item)` returns a raw payload.
- **SRC-2** Adapters emit a normalized `RawPost` — external id, author display name, author profile URL, body text, posted date, observed location hint, source name, fetch timestamp, raw payload blob.
- **SRC-3** Adapters declare their own rate budget and terms metadata; the ingest scheduler respects both.
- **SRC-4** Adding a source requires no change to extraction, scoring, storage or UI.
- **SRC-5** Each source can be enabled or disabled per tenant at runtime, so a source can be cut off in minutes without a deploy.
- **SRC-6** Sources are credential-isolated: one source's keys or session cannot be used by another.

## Ingestion pipeline

Ingestion's job is narrow: get candidate posts into raw storage, exactly once, with provenance intact, without hammering anyone. All interpretation happens downstream.

### Discovery

- **ING-1** A configurable pattern library drives discovery. Seed it with the phrasings people actually use: *excited to announce, thrilled to share, happy to share that I'm starting a new position, I've joined, joining the team at, new chapter, next chapter, first day at*, plus `#newjob`, `#newrole`, `#newbeginnings`, `#firstday`.
- **ING-2** LinkedIn's structured "started a new position at X" update is a distinct, higher-precision signal and is matched separately from free text.
- **ING-3** Negative patterns suppress obvious non-leads: `#opentowork`, *looking for my next role, we're hiring, congratulations to* (someone else's news), anniversary and promotion-without-move posts.
- **ING-4** Patterns are editable in the admin UI without a deploy, because phrasing drifts and each market has its own idiom.
- **ING-5** Discovery is scoped by market — a configured set of metros, each with a radius — so a run doesn't pull the whole country.

### Scheduling and politeness

- **ING-6** Runs are scheduled per source (default: every 6 hours) and are incremental via a stored cursor.
- **ING-7** Re-running a window is idempotent. The same post ingested twice produces one record.
- **ING-8** Each source has a configurable request budget per minute and per day. The scheduler enforces it; workers cannot exceed it.
- **ING-9** On `429` or `403`, the worker backs off exponentially. Three consecutive failures trip a circuit breaker that **halts the run and alerts an admin** rather than retrying harder or routing around the block.
- **ING-10** `robots.txt` and published API terms are honored for any web source.

## Storage and integrity

- **ING-11** The raw payload is stored verbatim alongside source name, source URL, fetch timestamp and the run id that produced it. Extraction is re-runnable against raw data without re-fetching.
- **ING-12** Exact deduplication on `(source, external_post_id)`. Near-duplicate detection on a normalized hash of author plus body, to catch cross-posts and re-shares.
- **ING-13** Person-level deduplication is separate and happens after extraction — the same person may be announced in a post and a press release.
- **ING-14** Items that fail parsing go to a dead-letter queue with the error, retryable from the admin UI. A failed item never silently disappears.

## Observability

- **ING-15** Every run records: source, start and end time, items seen, items new, items deduplicated, items errored, and the terminal status.
- **ING-16** An admin page shows the last 50 runs per source, and alerts fire on a run that returns zero new items where the prior three returned more than zero — the usual first sign that a source broke.

## Classification, extraction and enrichment

A raw post becomes a lead in three graded stages, cheapest filter first. Precision matters more than recall here: a wrong lead wastes an agent's phone call and erodes their trust in the whole feed, while a missed lead costs nothing visible.

### Stage 1 — Cheap gate

- **EXT-1** A regex and keyword prefilter drops anything with no plausible job-change signal, before any model is called. Target: discard 80% or more of raw volume at near-zero cost.

### Stage 2 — Classification

- **EXT-2** A fast LLM (Claude Haiku 4.5) returns `is_job_change: boolean` plus a 0–1 confidence for each survivor.

- **EXT-3** The classifier distinguishes a genuine job change from the common look-alikes: a promotion with no employer change, an anniversary, a congratulations post about someone else, a hiring post, a company milestone.
- **EXT-4** Posts below the confidence floor are dropped and logged, not queued — only extraction failures reach human review.

### Stage 3 — Structured extraction

- **EXT-5** A stronger model (Claude Sonnet 5) extracts to a strict JSON schema, returning `null` rather than guessing for any field not stated in the post:

| Field                         | Type    | Notes                                                                |
|-------------------------------|---------|----------------------------------------------------------------------|
| <code>person_name</code>      | string  | As written in the post                                               |
| <code>new_title</code>        | string  | Normalized against a title dictionary downstream                     |
| <code>new_company</code>      | string  | Resolved to a company record where possible                          |
| <code>new_location</code>     | object  | Raw string plus parsed city, state, country                          |
| <code>start_date</code>       | date    | Absolute, or resolved from "starting in March" against the post date |
| <code>prior_company</code>    | string  | Often present, drives relocation detection                           |
| <code>prior_location</code>   | string  | Often absent; fall back to profile location                          |
| <code>work_arrangement</code> | enum    | onsite, hybrid, remote, unknown                                      |
| <code>is_relocation</code>    | boolean | Derived, not asserted by the model alone                             |

- **EXT-6** Every field carries its own confidence, and the record carries an overall confidence.
- **EXT-7** Records below the overall threshold (default 0.75), or missing `new_location`, route to the human review queue instead of the live feed.
- **EXT-8** Extraction is versioned. Re-running a new prompt or model against stored raw posts backfills without re-fetching, and each lead records which extractor version produced it.

## Accuracy targets and regression

- **EXT-9** Maintain a golden set of at least 200 hand-labeled posts, including hard negatives. It is the gate for any prompt or model change.
- **EXT-10** Targets on the golden set: classifier precision  $\geq 0.95$  and recall  $\geq 0.80$ ; field accuracy  $\geq 0.90$  for title and company,  $\geq 0.85$  for location and start date.

## Enrichment

- **ENR-1** Geocode `new_location` to lat/lng through a provider-agnostic interface, with an aggressive cache keyed on the normalized location string.
- **ENR-2** Store geocode precision (city, metro, region) alongside the coordinates. **Resolution never goes finer than a city or metro centroid**, by policy — see the legal section.
- **ENR-3** Resolve `new_company` to a company record: name, domain, industry, size band, headquarters location.
- **ENR-4** Map `new_title` to a seniority band (IC, senior IC, manager, director, VP, C-level) using a maintained dictionary, not a model call.
- **ENR-5** Relocation is derived by comparing the new metro against the prior metro where known, falling back to the profile location. When neither is available, `is_relocation` is `unknown` — never defaulted to false.
- **ENR-6** An estimated compensation band may be derived from seniority, industry and metro. It is stored and displayed as an **estimate**, with its basis shown, and is never presented as fact.

## Human review

- **REV-1** The review queue shows the original post text beside the extracted fields, each field editable inline.
- **REV-2** A reviewer can approve, correct and approve, or reject with a reason.
- **REV-3** Corrections are logged as labeled training and regression data. The golden set grows from real corrections.

## Lead scoring

Score is a transparent weighted sum, 0–100, not a black box. An agent has to be able to read why a lead is at the top of their list, and — given the Fair Housing exposure — you have to be able to show a regulator the same thing.

| Signal                    | Weight | Source                                    | Why it predicts a transaction                                                             |
|---------------------------|--------|-------------------------------------------|-------------------------------------------------------------------------------------------|
| Relocation detected       | 30     | New metro ≠ prior metro                   | The single strongest signal. A cross-metro move nearly always means a housing transaction |
| Proximity to service area | 20     | Distance from the agent's configured area | A lead you cannot serve is worth zero regardless of quality                               |
| Start date within 90 days | 15     | Extracted start date                      | Housing decisions cluster in the weeks before and after a start date                      |
| Seniority band            | 15     | Title dictionary                          | Proxy for budget and for the ability to move quickly                                      |
| Post recency              | 10     | Post date                                 | Decays over 60 days. Week-old news beats month-old news                                   |
| Onsite or hybrid role     | 10     | Work arrangement                          | A fully remote role weakens the need to move at all                                       |

## Rules

- **SCO-1** Score bands: Hot  $\geq 70$ , Warm 40–69, Cool  $< 40$ . Bands drive map color and default sort.
- **SCO-2** Every lead stores its per-signal contributions. The card shows the top two as plain-language reasons ("Relocating to Austin", "Starts in 23 days").
- **SCO-3** Weights are configurable per tenant in settings. Changing weights rescores the existing corpus in the background and does not require a deploy.
- **SCO-4** Scores recompute on a schedule so recency decay stays accurate without a user action.
- **SCO-5** An unknown signal contributes zero — it never penalizes. A lead with a missing start date should not rank below a lead with a confirmed distant one.
- **SCO-6** Low extraction confidence caps the displayed score and marks the lead "unverified" rather than silently inflating it.

## Explicitly excluded from scoring

These are prohibited inputs, enforced at the schema level rather than by convention — the fields do not exist on the scoring input type:

- Any direct or inferred protected characteristic: race, color, religion, sex, familial status, national origin, disability.
- Proxies for them: name-derived ethnicity or gender, profile photograph, alma mater, surname frequency.
- Neighborhood-level demographic data of any kind.

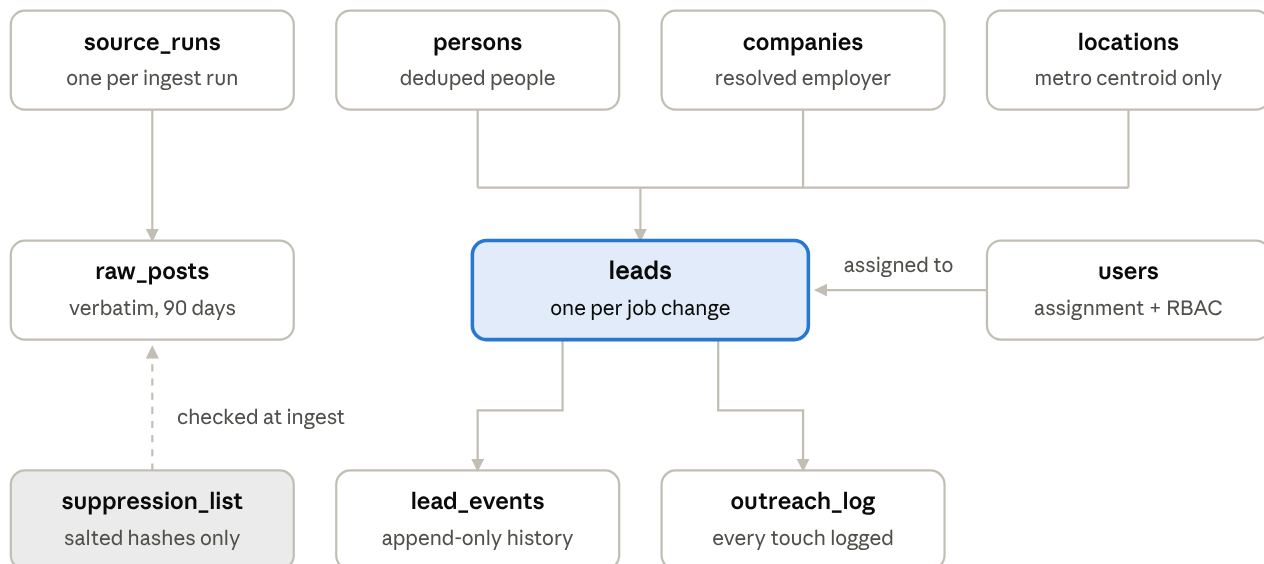
**SCO-7** Every scoring run writes its inputs and output to an audit log, retained for the life of the lead, so any ranking decision can be reconstructed after the fact.

## Data model and storage

Postgres with the PostGIS extension handles both halves of this workload — relational joins for the feed and spatial queries for the map — so there is no reason to split the store. The one structural rule that matters: **raw posts and derived leads are separate tables**, so you can re-extract without re-fetching and delete a person without losing your pipeline's audit trail.

### Every lead traces back to one raw post and one ingest run

Core entities only · the rest are listed under Supporting tables



core data model · 10 entities

## The leads table

| Column                | Type                   | Notes                                                           |
|-----------------------|------------------------|-----------------------------------------------------------------|
| id                    | uuid                   | Primary key                                                     |
| raw_post_id           | uuid                   | Provenance, nullable for CSV imports                            |
| person_id             | uuid                   | Deduplicated person across sources                              |
| new_company_id        | uuid                   | Resolved company                                                |
| new_title             | text                   | As extracted                                                    |
| seniority_band        | enum                   | Derived from the title dictionary                               |
| new_location_id       | uuid                   | Resolved location record                                        |
| geo                   | geography(Point, 4326) | City or metro centroid only                                     |
| geo_precision         | enum                   | city, metro, region                                             |
| prior_location_id     | uuid                   | Nullable                                                        |
| is_relocation         | enum                   | true, false, unknown — never a nullable boolean                 |
| start_date            | date                   | Nullable                                                        |
| work_arrangement      | enum                   | onsite, hybrid, remote, unknown                                 |
| score                 | smallint               | 0-100                                                           |
| score_factors         | jsonb                  | Per-signal contributions for the "why" display                  |
| extraction_confidence | numeric                | Drives the unverified flag                                      |
| extractor_version     | text                   | Which prompt and model produced this                            |
| status                | enum                   | new, contacted, nurturing, appointment, closed, lost, dismissed |
| assigned_user_id      | uuid                   | Nullable                                                        |
| detected_at           | timestampz             | When the system found it                                        |

| Column                  | Type                    | Notes                     |
|-------------------------|-------------------------|---------------------------|
| <code>posted_at</code>  | <code>timestampz</code> | When the person posted it |
| <code>deleted_at</code> | <code>timestampz</code> | Soft delete               |

## Supporting tables

`source_runs` and `raw_posts` hold acquisition and provenance. `persons`, `companies` and `locations` are the deduplicated dimension tables. `lead_events` is an append-only activity log per lead; `notes` and `outreach_log` record human touches. `users`, `teams` and `saved_views` cover access and personalization. `suppression_list`, `audit_log` and `extraction_labels` cover compliance and model quality.

## Indexes and constraints

- **DB-1** GiST index on `leads.geo` for viewport and radius queries.
- **DB-2** Composite btree on `(status, score DESC, detected_at DESC)` — the default feed query.
- **DB-3** Partial index excluding soft-deleted and dismissed rows, which are the common case to skip.
- **DB-4** GIN index on a tsvector of the post body for full-text search.
- **DB-5** Unique constraint on `(source, external_post_id)` in `raw_posts` — deduplication enforced by the database, not by application logic.
- **DB-6** All timestamps are `timestampz`. All geography is SRID 4326.

## Retention and deletion

- **DB-7** `raw_posts` are purged 90 days after ingestion. Extracted leads survive.
- **DB-8** Leads are soft-deleted by default; a hard-delete path exists for data subject requests and cascades to persons, enrichments, notes and events.
- **DB-9** A deleted person leaves a salted hash on the suppression list so the pipeline does not silently re-ingest them next week.
- **DB-10** Leads with no activity for 24 months are purged on a schedule.

## Dashboard — Leads feed

The default tab. A scrollable set of cards an agent can work top to bottom, where each card answers "who, where, and should I call them?" without opening anything.

### What a card shows

1. **Person** — name, and the headline as written.
2. **New role** — title at company, with the company logo and industry.
3. **New location** — city and state, with a **Relocating from 'city'** badge when a move is detected. This badge is the single most important element on the card.
4. **Start date** — absolute, plus a relative read ("starts in 23 days", "started 2 weeks ago").
5. **Score** — band chip plus the top two contributing reasons in plain language.
6. **Status** — current pipeline stage, changeable inline.
7. **Post excerpt** — two lines, with a link to the original.
8. **Quick actions** — assign, mark contacted, add note, dismiss.

### Requirements

- **FEED-1** Cards render in a responsive layout: one column on narrow screens, two to three on desktop, with consistent card height so the grid does not jump.
- **FEED-2** Default sort is score descending, then `detected_at` descending. Alternatives: newest first, start date soonest, nearest to my area.
- **FEED-3** Infinite scroll at 25 per page, with scroll position preserved when returning from a detail view.
- **FEED-4** Clicking a card opens the detail drawer over the feed. The feed does not navigate away or lose its position.
- **FEED-5** Multi-select with a checkbox, then bulk assign, bulk status change, bulk export, bulk dismiss.
- **FEED-6** Dismissed leads are hidden by default and recoverable through a filter — dismissal is never destructive.
- **FEED-7** Leads below the confidence threshold display an **Unverified** marker and, for admins, a one-click route into the review queue.
- **FEED-8** Distinct empty states: no leads yet, no leads matching these filters, and source error. Each names what to do next.
- **FEED-9** Skeleton cards while loading; no layout shift on arrival.

- **FEED-10** Keyboard navigable: **j** and **k** to move, **Enter** to open, **c** to mark contacted, **x** to dismiss.
- **FEED-11** New leads arriving since the page loaded surface as a **new leads** pill at the top rather than reordering under the agent's cursor.

## Dashboard — Map

The same result set as the feed, placed geographically. The question the map answers is one the card list cannot: *where is the activity clustering, and is any of it inside my farm area?*

- **MAP-1** One pin per lead at its new-location centroid. **City or metro resolution only** — the map must never imply a home address.
- **MAP-2** Clustering when pins overlap at the current zoom. A cluster shows its count and zooms to its bounds on click.
- **MAP-3** Pin color encodes score band by default, with a toggle to color by pipeline status instead. A legend is always visible.
- **MAP-4** Clicking a pin opens a compact preview card — name, new role, new location, score — with a link into the full detail drawer.
- **MAP-5** Leads sharing one centroid are jittered slightly and the pin carries a count badge, so a stack of eight people moving to the same city is visible rather than hidden under one marker.
- **MAP-6** Draw a radius or a freehand polygon to filter. **That spatial filter applies to the feed tab too** — the two tabs are two views of one query, never two separate queries.
- **MAP-7** A heatmap toggle replaces pins with density shading, for reading market-level patterns at low zoom.
- **MAP-8** The agent's configured service area renders as a persistent outlined overlay.
- **MAP-9** Filter state — spatial and otherwise — is encoded in the URL, so a view can be bookmarked and shared with a teammate.
- **MAP-10** Queries are bounded to the current viewport. Render 5,000 markers in under one second.
- **MAP-11** The map displays a reconciliation count when results differ from the feed: "412 leads, 389 mapped". The 23 unmappable ones are reachable in one click, not silently dropped.
- **MAP-12** The map is not the only route to any data. The feed tab is its accessible equivalent, which is what keeps this WCAG-conformant.

## Dashboard — supporting features

You asked not to overcrowd this, so the rule below is deliberate: **three tabs, and everything else is either chrome around them or tucked behind an admin role.** Most of what a dashboard like this accumulates belongs in a CRM, not here.

### Persistent chrome, shared by every tab

- **UI-1** A collapsible filter rail: market and radius, date range, score band, status, seniority, relocations only, company, industry, assigned-to, unverified-only. Filters apply to the feed and the map simultaneously.
- **UI-2** Saved views. Name a filter combination ("Hot relocations, Austin, this month") and pin it. Views are shareable within a team.
- **UI-3** A compact metrics strip above the tabs: **new leads this week, relocations this week, contacted rate, top inbound market.** Four numbers with a week-over-week delta. No charts here — if it needs a chart, it belongs in an export.
- **UI-4** Global search across person, company and title.
- **UI-5** A lead detail drawer, opened from anywhere: full post text, extracted fields with confidence, enrichment, map inset, activity timeline, notes, and the original post link.

### Third tab — Pipeline

- **UI-6** A simple kanban: New → Contacted → Nurturing → Appointment → Closed or Lost. Drag a card between columns; the move writes a `lead_event`.
- **UI-7** Column counts and a filter for "assigned to me". This is a working board, not a reporting tool.

### Admin only

- **UI-8** Review queue for low-confidence extractions, with a badge showing the backlog count.
- **UI-9** Source health: run history, error rates, the zero-results alert from the ingestion section.
- **UI-10** Settings: service area, score weights, pattern library, sources, team and roles, notifications.

### Deliberately left out

Named here so the decision is explicit and can be revisited rather than rediscovered:

- In-app email and calling sequences — the CRM does this better, and it drags TCPA and CAN-SPAM compliance into your product.
- Social-graph browsing and profile viewing — increases legal exposure for little lead value.
- AI-drafted outreach copy — tempting, and a v2 decision once the data quality is proven.
- Reporting and analytics beyond the four-number strip — export to a spreadsheet instead.
- Lead sharing or a marketplace between tenants — a data-rights problem, not a feature.

## Integrations and outreach

This system finds and ranks leads. It should hand off to whatever the agent already uses rather than becoming a second CRM they have to keep in sync.

- **INT-1** CSV export of the current filtered view, with every visible column plus the source URL.
- **INT-2** CRM push through a pluggable connector interface. Ship one connector first — Follow Up Boss is the usual choice for residential teams; HubSpot or Salesforce if the brokerage is larger.
- **INT-3** Push is idempotent and bidirectional on status: a lead marked Contacted in the CRM reflects back here, so the pipeline tab does not drift into fiction.
- **INT-4** A daily digest email: new leads in my service area above my score threshold, capped at ten, with a deep link into the filtered view.
- **INT-5** Optional Slack delivery of the same digest, plus an immediate alert for any Hot relocation inside the service area.
- **INT-6** An outbound webhook on lead created and lead status changed, for teams with their own automation.
- **INT-7** Every outreach touch is logged against the lead with channel, timestamp and user — whether initiated here or synced back from the CRM.
- **INT-8** The suppression list is checked before **any** export, push or digest. A suppressed person leaves the system and does not reappear in a CSV an hour later.
- **INT-9** Phone numbers, where present, are DNC-scrubbed before they are exposed for calling.

## Non-functional requirements

| Area                 | Target                                                                 | Why this number                                                                |
|----------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Lead corpus          | 100,000 leads, 1,000,000 raw posts                                     | Roughly 20 metros at a year of volume                                          |
| Concurrent users     | 50                                                                     | A mid-size brokerage, all logged in at 8am                                     |
| Feed query           | p95 under 500 ms                                                       | The agent scrolls; a stutter is felt immediately                               |
| Map viewport query   | p95 under 800 ms                                                       | Pan and zoom must not feel gated                                               |
| Initial page load    | Under 2.5 s to interactive                                             | Standard web vitals threshold                                                  |
| Freshness            | Post visible within 24 h, target 6 h                                   | A relocation lead goes stale in days, not hours                                |
| Availability         | 99.5% during business hours                                            | A read-mostly tool; overnight maintenance is fine                              |
| Classifier precision | $\geq 0.95$                                                            | A false lead costs an agent's call and their trust                             |
| Extraction accuracy  | $\geq 0.90$ title and company, $\geq 0.85$ location                    | Below this the review queue becomes the product                                |
| Running cost         | Under \$500/month at MVP scale                                         | Dominated by data licensing, not compute — confirm against the source decision |
| Browsers             | Evergreen Chrome, Safari, Firefox, Edge                                | No IE, no legacy mobile webviews                                               |
| Responsive           | Full function at tablet width; read-only is acceptable on phone for v1 | Agents triage on a laptop and check on a phone                                 |
| Accessibility        | WCAG 2.2 AA                                                            | The feed tab is the map's accessible equivalent                                |

**NFR-1** LLM cost scales with raw volume, not lead volume, which is why the Stage 1 cheap gate matters. At roughly \$0.002 per post for classification plus extraction, 50,000 posts a month is about \$100 — but without the prefilter it is several times that.

**NFR-2** All ingestion and extraction work is asynchronous. No user-facing request ever waits on a source fetch or a model call.

## Security, privacy and governance

The compliance machinery from the legal section becomes concrete here. Most of it is MVP scope because retrofitting deletion into a system that assumed permanence is a rewrite.

### Access

- **SEC-1** Authentication by SSO or email magic link. MFA required for admin roles.
- **SEC-2** Three roles: **agent** (sees leads in their assigned market), **manager** (sees the team, assigns leads, tunes scoring), **admin** (sources, settings, review queue, exports).
- **SEC-3** Multi-tenant isolation enforced at the database layer with row-level security, not by filtering in application code.
- **SEC-4** Export is a permissioned action, logged with the row count and the filter used.

### Data protection

- **SEC-5** TLS in transit, encryption at rest, and column-level encryption for contact details.
- **SEC-6** Secrets live in a managed secret store. No credentials in the repository, in environment files, or in the source configuration table in plaintext.
- **SEC-7** Source credentials are scoped per tenant and per source, rotatable without downtime.

### Privacy rights

- **PRIV-1** Lookup by name or profile URL returns everything held about a person, exportable in a machine-readable format.
- **PRIV-2** Deletion on request completes within 30 days and cascades to derived records, enrichments, notes and events.

- **PRIV-3** Deletion flags any CRM the person was pushed to, so the obligation is visible rather than quietly unmet downstream.
- **PRIV-4** A public-facing privacy notice states what is collected, from where, why, and how to request deletion.
- **PRIV-5** An opt-out request adds a salted hash to the suppression list, which is checked at ingestion — so suppression survives deletion without retaining the person's data.
- **PRIV-6** The data is used for the tenant's own outreach only. No resale, no sublicensing, no cross-tenant pooling. State this in the terms of service and enforce it in the schema.

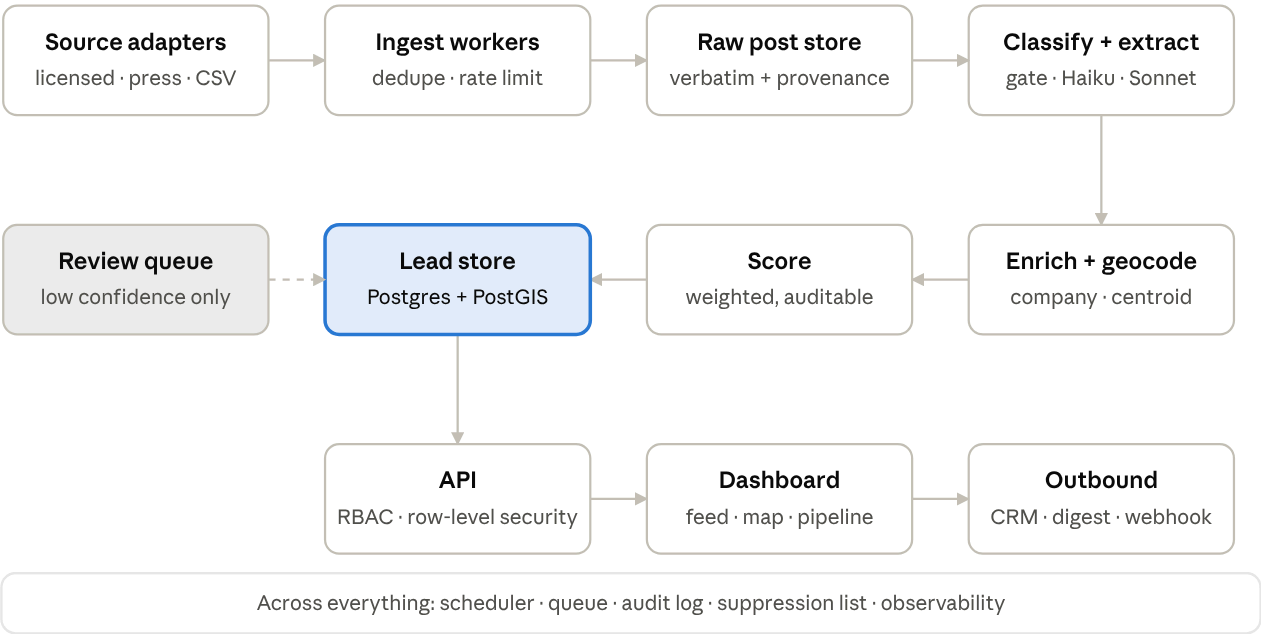
## Audit

- **AUD-1** Append-only audit log of who viewed, exported, contacted, edited or deleted which lead, and when.
- **AUD-2** Scoring inputs and outputs are retained for the life of the lead — the Fair Housing requirement from the scoring section.
- **AUD-3** Every lead can show its full provenance chain on demand: source, URL, fetch time, extractor version, score version.

# Architecture and recommended stack

Raw posts are stored before extraction, so leads rebuild without refetching

Pipeline left to right; the two stores are the seam between acquisition and interpretation



system architecture · 11 components, two stores

The seam that matters is between the raw post store and extraction. Everything left of it is acquisition, which is the part most likely to be replaced; everything right of it is interpretation, which you can re-run against stored data whenever the prompts or the model improve.

| Layer           | Pick                                                      | Why this one                                                                                                  |
|-----------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Database        | Postgres 16 + PostGIS                                     | Relational joins for the feed and spatial queries for the map, in one store. Managed on Neon, Supabase or RDS |
| App and API     | Next.js App Router + tRPC + Drizzle                       | One language end to end for a small team; typed from schema to card                                           |
| Background work | A durable job runner (Inngest, or Postgres-backed queues) | Ingestion and extraction must survive restarts and retry cleanly                                              |

| Layer      | Pick                                                                     | Why this one                                                                    |
|------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Extraction | Claude Haiku 4.5 for the gate, Claude Sonnet 5 for structured extraction | Two tiers because the gate runs on everything and extraction runs on a fraction |
| Maps       | MapLibre GL JS with MapTiler or Protomaps tiles                          | Avoids per-load pricing that punishes exactly the usage you want                |
| Clustering | Supercluster                                                             | Handles the 5,000-marker target client-side without a custom tile layer         |
| Geocoding  | Provider-agnostic interface, cached                                      | The cache matters more than the vendor; most locations repeat                   |
| Hosting    | Vercel plus a managed Postgres, or Fly.io for both                       | Either is fine at this scale. Do not build infrastructure yet                   |

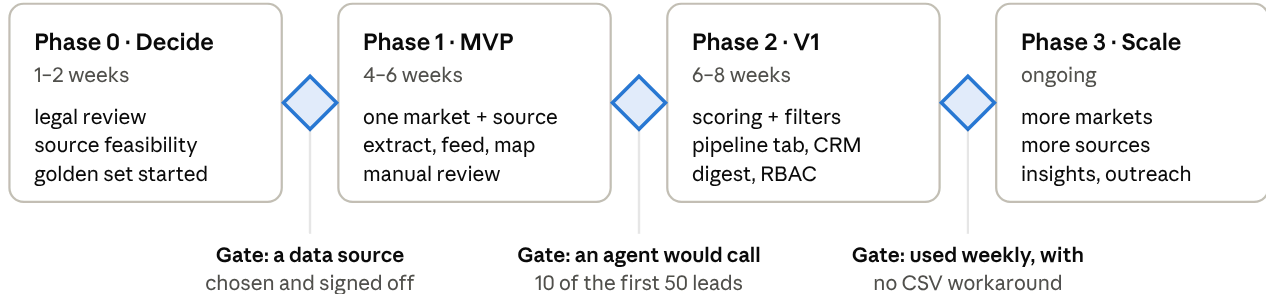
## Deliberate non-choices

- **No custom NER model in v1.** LLM extraction against a strict schema reaches useful accuracy in days, not months. Revisit only if volume makes per-post cost dominate.
- **No document database.** This workload is relational and spatial, and Postgres does both. A second store is a sync problem you do not need.
- **No Elasticsearch in v1.** Postgres full-text search is sufficient well past 100,000 leads.
- **No microservices.** One deployable with background workers. The ingestion and extraction split is a module boundary, not a network boundary.

# Phased delivery plan

**Each phase has one exit criterion; the source decision gates all of it**

Equal bands, not to scale · durations are estimates for a small team



delivery phases · 4 phases, 3 gates

The gates matter more than the durations. Each one is a question you can answer yes or no without a meeting, and none of them is "the code is finished".

**Phase 0 is not optional and is not a formality.** The source decision determines the legal posture, the monthly cost and the achievable volume — and it is the only decision here that is cheap now and expensive in six months. Build the CSV import path during this phase so the rest of the system has something to run against before a source is live.

**Phase 1 deliberately ships without scoring.** Sort by recency, show every lead, and have a human read them. You cannot design a scoring model before you have seen what the extracted data actually looks like, and the first 200 leads are also the golden set.

**Phase 2 is where it becomes a product rather than a list.** Scoring, saved views and the CRM connector are what turn a daily look into a daily workflow.

**Phase 3 is deliberately vague.** What to build after v1 depends on what the first paying team actually does with it, and committing to it now is guessing.

## Open questions

Six answers that change the build. The first two change it most.

1. **Residential or commercial?** The entire lead profile differs. A commercial version cares about company relocations and headcount growth, not individuals announcing a move. This document assumes residential.

2. **Who is the customer — one agent, one team, or brokerages as a SaaS product?** Multi-tenancy, RBAC, billing and the data-rights story all hinge on this. The document assumes a team, which is the middle case and the safest to build toward.
3. **What is the appetite for licensing data versus scraping, and what is the monthly budget for it?** This is the source decision from the legal section, and it is the one worth settling first.
4. **Which market launches first?** One metro is enough to prove the thesis and keeps geocoding and enrichment costs trivial.
5. **What CRM is already in use?** It determines which connector ships in v1 and whether the pipeline tab is even needed.
6. **Does outreach happen in this product or does it hand off?** The document assumes hand-off. Bringing outreach in-house adds TCPA and CAN-SPAM scope that roughly doubles the compliance surface.